Shreyas Govinda Raju

Backend, Data & ML Platform Engineer

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Summary

Software engineer with **8+ years at Amazon** building distributed backend services, ML infrastructure, and Amazon's public open-source developer SDKs. Drives technical strategy for cross-functional platform initiatives owning architecture from design through production at global scale, root-causing performance regressions that logs and dashboards cannot explain, and multiplying team output through cross-team design reviews and mentorship.

Technical Skills

Languages Python, Java, TypeScript, JavaScript, SQL, Bash

Distributed Systems Microservices, event-driven architecture, asynchronous processing, API design, concurrency, rate limiting, observability

Cloud & Infrastructure AWS (ECS / Fargate, Lambda, SQS, Step Functions, S3, DynamoDB, IAM), Docker, AWS CDK, CI/CD

ML & Data PyTorch, Apache Spark / PySpark, XGBoost, Amazon Redshift, AWS Glue

APIs & Tools GraphQL, Git, Swagger Code generation

Experience

Amazon

Seattle, Washington

Software Development Engineer 2 - Machine Learning & Data Platform — *May 2024 - Current*

- Spearheaded the architecture and delivery of a production **ML inference platform** public API, request-state store, distributed work queue, and auto-scaling model-serving fleet launching in 5 months and reliably handling peak loads of **50 TPS** for compute-intensive inferences.
- Architected high-throughput data pipelines to productionize multiple batch ML models, processing **1 billion records daily** to serve highly available online and offline customer predictions.
- Reduced inference p90 latency by **>95%**, serving infrastructure by **20x** (60 CPU nodes to 3 GPU nodes), and serving compute cost by **90%** by building the business case for a CPU-to-GPU migration and securing org buy-in before committing spend.
- Drove engineering standards across the organization led **cross-team architecture and design reviews**, established the platform's observability and on-call practices, and **mentored engineers and interns** to elevate team execution.
- Root-caused a fleet-wide latency regression that logs and dashboards could not explain isolated the variable using a control model on identical hosts and shipped startup diagnostics to identify processor-generation differences (**AVX-512 vs AVX2**), resolving inconsistent inference performance.
- Scaled an ML scoring platform from **1 to 5 regions** across 10+ international marketplaces, owning orchestration, feature-generation, and **PyTorch** model-training pipelines supporting new-product discovery.
- Reduced new-marketplace onboarding from weeks to a configuration change by decomposing a monolithic training pipeline into modular, **configuration-driven workflows** with per-marketplace and per-model overrides.
- Led standardization of **Spark** job packaging and deployment across 8 production pipelines, aligning owners across multiple teams and authoring the migration guide adopted organization-wide.

Software Development Engineer 2 - Commerce Platform & Partner Integrations — *Jun 2022 - May 2024*

- Spearheaded the architecture of a centralized **event-processing framework** event subscriptions, rate-limited queues, pluggable handlers, and output adapters behind a common interface accelerating partner time-to-market by cutting integration development from weeks to a single configuration.
- Built the framework's **data-enrichment layer** a typed API client over Order, Inventory and Availability **GraphQL** queries with secure credential management, converting sparse event notifications into complete business objects.
- Shielded downstream systems from traffic bursts by implementing **concurrency controls** across the queue-to-compute path and exposing queue-depth and dead-letter monitoring directly to integration owners.
- Automated partner-facing **API documentation** from the live schema, replacing a manual process and preventing internal-only fields from being published externally.

Software Development Engineer - Alexa Skills Kit, Developer Tools & Public SDKs — *Jul 2018 - Jun 2022*

- Core contributor to Amazon's official open-source **Alexa Skills Kit SDKs** for Python, Node.js and Java and the ASK CLI published to **PyPI, npm and Maven** and consumed by third-party developers building on the Alexa platform.
- Owned the **API specification and code-generation pipeline** for three language SDKs, where a single API change propagated across every published client library; enforced release boundaries to prevent unreleased functionality from reaching public SDKs.
- Designed a **declarative dialog framework** for the Node.js SDK, replacing hand-written state machines with a composable control tree supporting initiative, confirmation, and validation.
- Built the public **Skill Management API SDK for Python** from scratch, including authentication, response modeling, testing, and packaging for public and internal distribution.
- Shipped core **ASK CLI** capabilities across two major versions, including project scaffolding, extensible build hooks, API commands, model-accuracy profiling, and local debugging for live requests.

Education

University of Southern California, Los Angeles

Aug 2016 - May 2018

M.S. in Computer Science

National Institute of Technology Karnataka, Surathkal, India

Jul 2011 - May 2015

B.Tech in Computer Science and Engineering